5

The Direction of Context Effects

What Determines Assimilation or Contrast in Attitude Measurement?

In Chapter Four we reviewed a number of different sources of context effects in attitude measurement. In the present chapter, we focus in more detail on one particular source, namely, substantive information that is brought to mind by preceding questions. This information may affect the judgment stage of the response answering process and result in assimilation or in contrast effects. To conceptualize the underlying processes, we draw on the inclusion/exclusion model proposed by Schwarz and Bless (1992a). This model specifies the conditions under which question order effects emerge at the judgment stage and predicts their direction, size, and generalization across related issues.

Although the model has received a fair amount of empirical support in psychological experimentation (see again Schwarz & Bless, 1992a, for a review), we emphasize at the outset that its predictions go far beyond the data given. We consider this an advantage rather than a shortcoming. Traditionally, research on context effects in surveys has been hampered by a lack of theoretical frameworks and has been characterized by a more or less ad hoc nature. In this as in other areas of research, however, we are likely learn useful lessons from a comprehensive model, even if empirical work eventually proves parts of it wrong.

The Inclusion/Exclusion Model

The key assumptions of the inclusion/exclusion model draw heavily on recent research in cognitive psychology that emphasizes the dynamic nature of knowledge representations (see Barsalou, 1989; Kahneman & Miller, 1986; and the contributions in Martin & Tesser, 1992). Moreover, the model's assumptions and predictions regarding the emergence of assimilation effects are compatible with the belief sampling model proposed by Tourangeau and colleagues (Tourangeau, 1987, 1992; Tourangeau, Rasinski, & Bradburn 1992). In contrast to the belief sampling model, however, the inclusion/exclusion model explicitly identifies the conditions that give rise to contrast effects. For that reason, our discussion in this chapter draws on the conceptualization offered by the inclusion/exclusion model, although this model is largely identical with the belief sampling model of Tourangeau, Rasinski, & Bradburn (1992) with regard to the conceptualization of assimilation effects and the stability of attitude reports over time.

Constructing Targets and Standards

Schwarz and Bless (1992a) assume that when individuals are asked to form a judgment about some target stimulus they first must retrieve some cognitive representation of it. In addition, they need to determine some standard of comparison to evaluate the stimulus. Both the representation of the target stimulus and the representation of the standard are, in part, context-dependent. Individuals do not retrieve all knowledge that may bear on the stimulus, nor do they retrieve and use all knowledge that may potentially be relevant to constructing a standard. Rather, they rely on the subset of potentially relevant information that is most accessible at the time of judgment (Bodenhausen & Wyer, 1987; Higgins, 1989; Higgins & King, 1981). Accordingly, their temporary representation of the target stimulus as well as their construction of a standard includes information that is chronically accessible—hence, context-independent—as well as information that is only temporarily accessible due to contextual influences.

Whereas differences in the chronic accessibility of information reflect respondent characteristics, differences in the temporary accessibility of information are primarily due to questionnaire variables. Most important, information that has been used for answering a preceding question is particularly likely to come to mind when respondents are later asked a related question. How the information that comes to mind influences the judgment depends on whether it is used to construct a representation of the target, a representation of a standard or scale anchor against which the target is evaluated or whether it is excluded from all of these.

Assimilation Effects

Information that is included in the temporary representation that individuals form of the target of judgment will result in assimilation effects because the judgment is based on the information included in the representation used. Accordingly, the addition of information with positive implications results in a more positive judgment, whereas the addition of information with negative implications results in a more negative judgment.

The size of context-induced assimilation effects increases with the amount and extremity of the temporarily accessible information and decreases with the amount, extremity, and evaluative consistency of chronically accessible information included in the representation of the target. Hence, the size of assimilation effects is a function of context as well as of respondent variables. On the one hand, the size of assimilation effects increases the more, or the more extreme, information the preceding question brings to mind. On the other hand, the impact of the temporarily accessible information is limited by the chronically accessible information. For example, adding positive information to a representation that includes only one piece of negative information will have more impact than adding it to a representation that includes numerous negative attributes. By the same token, adding a piece of positive information to a representation that includes, for example, two positive and two negative attributes will have more impact than adding it to a representation that includes four negative attributes. Thus, the size of assimilation effects decreases as more chronically accessible information and more evaluatively consistent information is included in the representation of the target.

This conceptualization has important implications for the impact of respondents' expertise on the emergence of context effects. In addition, it bears on how one versus several preceding questions affect the size of context effects. We address each of these issues in turn.

Expertise. Survey researchers have often assumed that respondents who are highly knowledgeable about an issue should be less susceptible to context effects than those who are less knowledgeable, but relevant empirical findings are mixed (see Bickart, 1992; Schuman & Presser, 1981). The present model allows us to specify the crucial conditions in more detail. First, let us suppose that a respondent has a large amount of evaluatively consistent information about target X that is chronically accessible in memory. Increasing the temporary accessibility of some additional information about X through preceding questions should have little impact on the respondent's evaluation of X. This is the case in which the usual assumption of experts' resistance to context effects should hold. But if the expert's chronically accessible information is evaluatively mixed, the temporarily accessible information is still likely to make a difference. Hence, the evaluative consistency of the expert's chronically accessible information is the crucial boundary condition.

As a second possibility, let us suppose that the respondent's expertise pertains to the general content domain rather than to the specific target addressed in the question of interest. For example, a market research survey could ask respondents to evaluate a new car with which they have not yet had any experience. If the survey draws attention to price, this information would probably have more meaning for respondents who know more about cars in general (see Herr, 1989). For example, experts could infer features that a car in the $40,000 price range is likely to have whereas the same information would hold less meaning for novices. In terms of our model, a question pertaining to the adequacy of the $40,000 price tag would render more information temporarily accessible for experts than for novices. As a result, this question should produce stronger context effects among experts than among novices. Thus, the crucial issue is what respondents' expertise pertains to: if they are experts on the specific car under study we would not expect them to show strong context effects. In the present example, however, respondents are experts on the general domain (cars), but not on the specific target (this particular car) and thus may bring more domain knowledge to bear on the target, allowing for the emergence of stronger context effects.

In general terms, these considerations predict that experts on target X are less likely to show strong context effects in judgments of X than novices provided that their chronically accessible information is evaluatively consistent. But if information about X is relevant for evaluating target Y, experts on X should show stronger context effects in their judgment of Y than novices, reflecting that questions about X bring more information to the experts' mind. Because most studies have only assessed respondents' general domain expertise without considering the relationship of this expertise to the target under study, it is not surprising that the available findings are mixed.

As this discussion illustrates, how we define expertise is of crucial importance. On theoretical grounds, expertise may pertain to the general content domain or to the specific issue under study. Both forms of expertise have different implications for the emergence of context effects. Unfortunately, many discussions of respondents' expertise not only ignore this distinction but also draw solely on global proxy variables, such as years of schooling or self-reported interest. Such variables are unlikely to moderate the size of context

effects unless they are confounded with the actual amount of knowledge about the target issue. It should come as little surprise then that formal education or self-reported interest have been found to show inconsistent relationships with the emergence and size of context effects (see Bradburn, 1983; Schuman & Presser, 1981).

Number of Preceding Questions. Whereas our discussion of expertise on target issue X pertained to the amount of chronically accessible information about it, similar considerations hold for the amount of information rendered temporarily accessible by preceding questions. Specifically, the impact of a piece of information brought to mind by one question is reduced as additional information is brought to mind by other questions. Hence, the impact of a given question decreases as the number of related context questions increases. For example, answering a question about marital happiness had a pronounced impact on subsequent reports of general life satisfaction when respondents' marriage was the only specific domain addressed. This impact was significantly reduced, however, when respondents had to answer questions about their leisure time and their job in addition to questions about their marriage before reporting their general life satisfaction (Schwarz, Strack, & Mai, 1991). In addition, the increased accessibility of information used to answer a preceding question wears off over time, further reducing its impact on questions that are separated by a sufficient number of intervening items (see Tourangeau & Rasinski, 1988; Tourangeau, 1992).

Contrast Effects

According to the inclusion/exclusion model, the same piece of information that elicits an assimilation effect may also result in a contrast effect. The inclusion/exclusion model (Schwarz & Bless, 1992a) holds that a contrast effect occurs when the information is excluded from rather than included in the cognitive representation formed of the target.

For example, let us suppose that a given piece of information with positive (or negative) implications is excluded from the representation of the target category. In this case, the representation will contain less positive (or negative) information, resulting in less positive (or negative) judgments. We call this possibility a *subtraction-based contrast effect* (see Bradburn, 1983; Schuman & Presser, 1981, for a related discussion). The size of subtraction-based contrast effects increases with the amount and extremity of the temporarily accessible information excluded from the representation of the target, and it decreases with the amount, extremity, and evaluative consistency of the information that remains in the representation of the target, paralleling our discussion of assimilation effects. Hence, we would again expect experts to show less pronounced subtraction-based contrast effects because they have a larger amount of chronically accessible information to use in constructing the representation of the target.

However, respondents may not only exclude accessible information from the representation formed of the target but also may use this information in constructing a standard or scale anchor. If the implications of the temporarily accessible information are more extreme than the implications of the chronically accessible information used in this construction, the process will result in a more extreme standard and elicit contrast effects for that reason. The size of such *comparison-based* contrast effects increases with the extremity and amount of temporarily accessible information used in constructing the standard or scale anchor and decreases with the amount and extremity of chronically accessible information used in making this construction.

Which of these processes drives the emergence of a contrast effect determines whether the contrast effect is limited to a single target or generalizes across related targets. If the contrast effect is based on the mere subtraction of information from the representation formed of the target, it is limited to the evaluation of that particular target. This simply means that the evaluation is based on the information "left" in the representation of the target. But if the

information excluded from the representation of the target is used in constructing a standard or scale anchor, contrast effects are likely to emerge on each judgment to which this standard or scale anchor is relevant.

Because the model provides two related mechanisms for the emergence of contrast effects, a logical question is raised. Under which conditions are we likely to obtain subtraction-based effects and under which are we likely to obtain comparison-based effects? The model holds that information that is excluded from the representation of the target is only used in constructing a standard or scale anchor if it has been thought about with regard to the relevant judgmental dimension. In Chapter Four, we reviewed a study on scale anchoring by Schwarz, Münkel, and Hippler (1990) that illustrates this point (see Table 4.2). Respondents were asked to rate a number of beverages according to how "typically German" they were. When this question was preceded by a question about the frequency with which Germans drink beer or vodka, contrast effects emerged in the typicality ratings. But when the preceding question pertained to the caloric content of beer or vodka, the typicality ratings were unaffected. The pattern of findings indicated that respondents only used the highly accessible drinks information in constructing a standard or scale anchor when the question that brought the drinks to mind addressed the underlying dimension of frequency of consumption that was relevant to a typicality judgment.

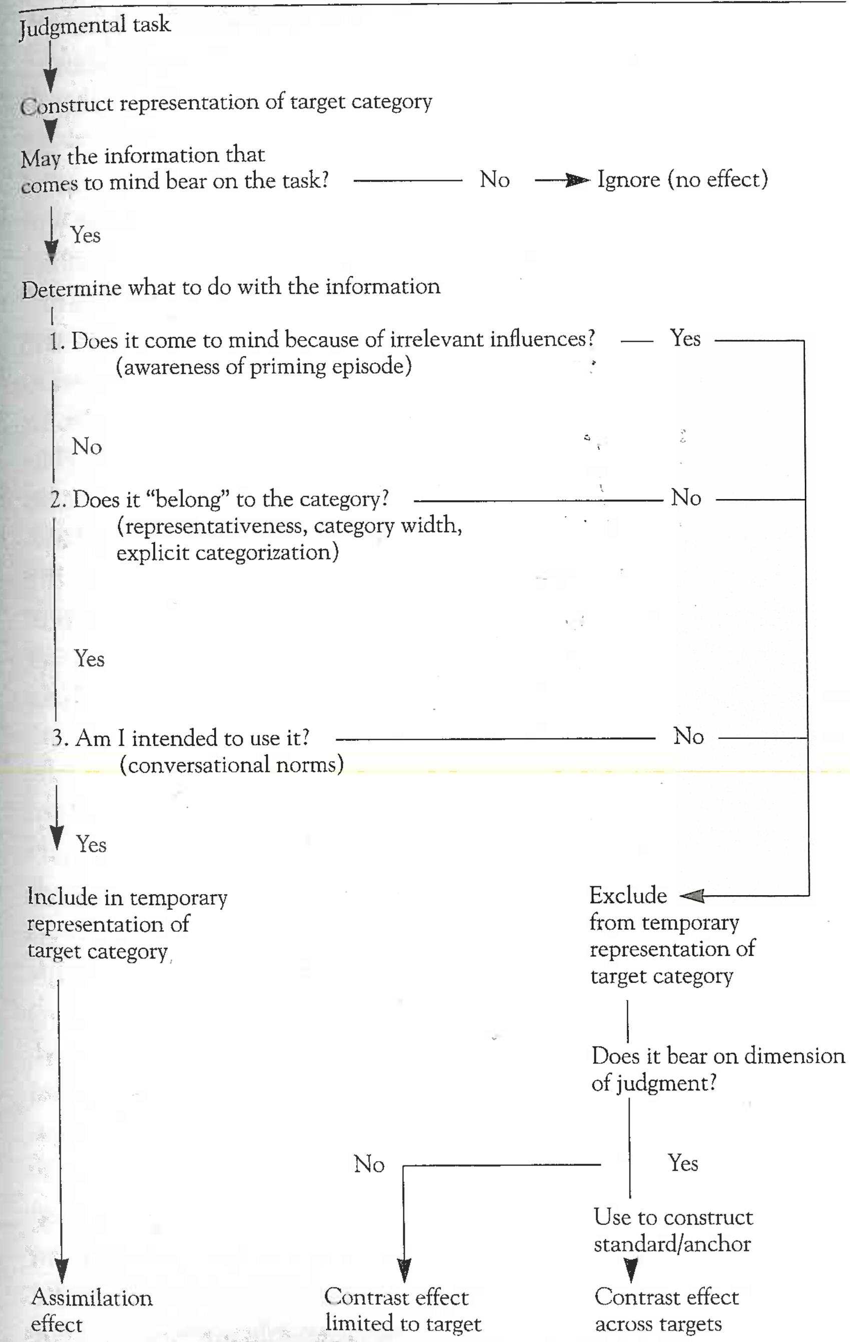
In sum, the inclusion/exclusion model holds that information that is excluded from the representation of the target category results in subtraction-based contrast effects if it has not been thought about with regard to the underlying dimension of judgment. If it has been thought about with regard to the relevant dimension, the excluded information is likely to be used in constructing a standard or scale anchor, resulting in comparison-based contrast effects. Whereas subtraction effects are limited to the evaluation of the target from which the information is excluded, comparison-based contrast effects generalize to the evaluation of every target to which the standard or scale anchor is relevant.

What Triggers the Exclusion of Information?

We assume that the default operation for most respondents is to include relevant information that comes to mind in the representation of the target. In other words, we should be more likely to see assimilation effects than contrast effects in survey research, an issue that should be addressed by meta-analyses. In contrast, the exclusion of information needs to be triggered by salient features of the question answering process. In principle, any variable that affects the categorization of information is likely to affect the emergence of assimilation and contrast effects, linking the present model to cognitive research on categorization processes in general (see Barsalou, 1989; Smith, 1990, for reviews). Schwarz and Bless (1992a) review a host of heterogeneous variables that have been shown to affect context effects in social judgment and trace their operation to inclusion/exclusion processes. Here we address the variables that are of particular relevance to survey research. The variables can be conceptualized as bearing on three decisions that respondents have to make with regard to the information that comes to mind, as shown in Figure 5.1.

Some of the information that comes to mind may simply be irrelevant, pertaining to issues that are unrelated to the question asked. Other information may be relevant and respondents have to decide what to do with it. The first decision relates to *why* this information comes to mind. Information that seems to come to mind for the "wrong reason"—for example, respondents are aware of the potential influence of a preceding question—is likely to be excluded (Lombardi, Higgins, & Bargh, 1987; Ottati, Riggle, Wyer, Schwarz, & Kuklinski, 1989; Strack, Schwarz, Bless, Kübler, & Wänke, 1993). The second decision relates to whether the information that comes to mind *belongs to* the target category. The content of preceding questions (Schwarz & Bless, 1992a), the width of the target category (Schwarz & Bless, 1992b), the extremity of the information (Herr, 1986), or its representativeness for the target category (Strack, Schwarz, & Gschneidinger, 1985) are relevant at

FIGURE 5.1 Emergence of assimilation and contrast effects.



this stage. Finally, conversational norms may determine respondents' perception of what they are supposed to do with highly accessible information (Schwarz, Strack, & Mai, 1991; Strack, Martin, & Schwarz, 1988; Tourangeau, 1984; Tourangeau, Rasinski, & Bradburn, 1992).

Whenever any of these decisions results in the exclusion of information from the representation formed of the target, it will elicit a contrast effect, the size of which will depend on the variables discussed earlier. Whether the contrast effect is limited to the target or generalizes across related targets depends on whether the excluded information is merely subtracted from the representation of the target or used in constructing a standard or scale anchor. In contrast, when the information that comes to mind is included in the representation formed of the target, it results in an assimilation effect, the size of which also depends on the variables discussed earlier. Hence, the model predicts the emergence, the direction, the size, and the generalization of context effects in attitude measurement. In addition, it predicts the stability of attitude reports over time and offers some predictions about the dependence of these effects on the mode of data collection used.

Stability Over Time

The model's predictions about the stability of attitude judgments over time follow directly from the assumptions about the construction of representations of targets and standards (see Tourangeau et al., 1992, for a related discussion). Specifically, repeating the judgment process at different points in time will result in similar judgments to the extent that respondents form similar mental representations of the target and of the standard each time. Several variables determine how likely this is to be the case.

Obviously, no change is expected when the context of the attitude judgment remains the same, rendering the same information temporarily accessible. Similarly, no change is expected when the judgment is based solely on chronically accessible information that

comes to mind at both points in time. Findings reported by Tourangeau et al. (1992) provide some support for this conclusion. They assessed respondents' attitudes toward abortion by asking them to report their thoughts in an open-ended format. When this procedure was repeated three weeks later, respondents' global attitude report was more strongly correlated with their previous attitude report when their open-ended responses at *t2* showed overlap with their thoughts at *t1* than when they did not. For example, when respondents reported two identical thoughts at both times, their judgments correlated $r = .84$ whereas this correlation was only $r = .35$ when their reported thoughts showed no overlap.

Even under conditions where the mental representations formed at both times include considerable amounts of different information, these differences in representation will only result in different judgments if the information used at *t1* and *t2* has different evaluative implications. Simply replacing one piece of information with a different one of similar valence will not change the evaluative judgment.

Our discussion of the size of context effects also bears directly on stability over time. As noted earlier, the size of assimilation effects decreases as the amount and evaluative consistency of other information included in the representation of the target increases. Hence, adding a piece of information at *t2* to a representation of the target that is otherwise identical with that used at *t1* will only result in change if the initial representation was based on a small amount of information or was evaluatively inconsistent. Support for the role of evaluative consistency is again provided by the Tourangeau et al. (1992) study. In their reinterview study, attitude measures assessed three weeks apart were correlated $r = .57$ for respondents who reported having mixed views on the topic, but $r =$ 81 for respondents who reported being clearly on one side of the issue.

The same logic applies to changes in the representation of the target if a piece of information used at *t1* is deleted at *t2*. Deleting a piece of previously used information will only result in different judgments if little other information was included in the first place or if the representation was evaluatively inconsistent.

Finally, similar considerations apply to changes in the representation of the standard, again paralleling our discussion of the size of comparison-based contrast effects. Accordingly, the variables that determine the size of context effects are also the variables that determine the stability of evaluative judgments over time.

Mode Effects

As already noted, the inclusion/exclusion model assumes that the default operation is to include information that comes to mind whereas the exclusion of information needs to be triggered by salient features of the question answering process. Moreover, exclusion processes require additional cognitive work. Accordingly, we should only see them when respondents are motivated and able to engage in this extra work. Consistent with this assumption, Martin, Seta, and Crelia (1990) observed that a lack of motivation—or distracting subjects from engaging in cognitive work—eliminated the emergence of contrast effects that were otherwise obtained. This suggests that we should be more likely to see contrast effects if respondents have a greater chance to engage in the necessary cognitive work. For example, such effects should be more likely under the leisurely conditions of a self-administered questionnaire (provided that respondents do not read all questions before answering them, thus eliminating question order; for example, Schwarz & Hippler, 1995) than under the time pressure of a telephone interview. However, data bearing on this assumed impact of the differential pace of different modes of data collection are not yet available.

Implications for Questionnaire Construction

How do the model's assumptions bear on the emergence of assimilation and contrast effects in survey research? In this section of the chapter, we review a number of different questionnaire variables,

reporting empirical evidence where available. The most important variables are the content and number of preceding questions, the generality of the target question, the spacing of related questions in the questionnaire, introductions or the lack of introductions to a block of questions, and the graphical presentation of questions in self-administered questionnaires.

The Content of Preceding Questions

The content of preceding questions determines the information that becomes temporarily accessible in memory. In addition, it may determine respondents' decisions about whether the information brought to mind belongs or does not belong to the target category to be evaluated (see Schwarz & Bless, 1992a; Tourangeau, Rasinski, & D'Andrade, 1991).

In one of their studies, Schwarz and Bless (1992a) asked a sample of German college students to evaluate the Christian Democratic Party (CDU) in power in Germany. To do so, respondents presumably recalled chronically accessible information from memory, which may have included that the CDU was a conservative party, that Chancellor Kohl was a member of it, and so on. For some respondents, the party evaluation question was preceded by one of two political knowledge questions, each of which pertained to a specific politician, namely Richard von Weizsäcker. This politician, who was a member of the CDU, was highly respected by Germans no matter their party preference. To elicit the inclusion of Richard von Weizsäcker in respondents' representation of the Christian Democrats, Schwarz and Bless asked some respondents, "Do you happen to know which party Richard von Weizsäcker has been a member of for more than twenty years?" (See Table 5.1.)

As shown in the first row of Table 5.1, answering this question resulted in more favorable evaluations of CDU politicians than a control condition in which no question about Richard von Weizsäcker was asked. This assimilation effect reflects the inclusion of Richard von Weizsäcker in the representation formed of the CDU.

TABLE 5.1 Evaluation of political parties with and without inclusion of a respected politician.

	Preceding question about Richard von Weizsäcker:		
	Relating to his party membership	*No question*	*Relating to his presidency*
Target			
Christian Democrats	6.5	5.2	3.4
Social Democrats	6.3	6.3	6.2

Source: Adapted from Schwarz and Bless, 1992a.

Note: N = 19 to 25 per condition; 1 = unfavorable and 11 = very favorable opinion about politicians of the respective party.

However, Richard von Weizsäcker had not only been a member of the CDU for several decades. At the time of the study, he was also serving as president of the Federal Republic of Germany, and the office of president required him to refrain from active participation in party politics. As a representative of the Federal Republic, the president is supposed to take a neutral stand on party issues, much like the Queen in the United Kingdom. This position rendered him particularly suitable for the present experiment because it allowed the authors to ask other respondents, "Do you happen to know which office Richard von Weizsäcker holds, setting him aside from party politics?" Answering this question was expected to *exclude* Richard von Weizsäcker from the representation that respondents formed of CDU politicians, resulting in a contrast effect. As shown in the first row of Table 5.1, this prediction was again confirmed in comparison to the control condition.

Since neither the party membership question nor the presidency question tapped the evaluative dimension, the model predicts that the observed contrast effect would reflect a mere subtraction process. According to this assumption, Richard von Weizsäcker was chronically accessible to some respondents in the control condition. Answering the party membership question, however, rendered him

accessible for all respondents, resulting in a more positive average evaluation of the party. In contrast, answering the presidency question induced all respondents, including those for whom he was chronically accessible, to exclude him from the representation formed of his party. As a result, the party membership question increased and the presidency question decreased the number of respondents who included this highly respected politician in their representation of the CDU. The result was the observed differences in judgment. The obtained contrast effect, however, should have been limited to evaluations of the Christian Democrats and not have generalized to evaluations of related targets. And indeed, evaluations of the Social Democrats, shown in the second row of Table 5.1, were not affected by the context questions about Richard von Weizsäcker. But had the context questions tapped the evaluative dimension, the model would predict that the contrast effects would have generalized across targets because Weizsäcker would be used in constructing a standard.

In summary, this study illustrates that the same information (in the present case, information about a highly respected politician) may result in assimilation or contrast effects in judgment depending on whether it is included in or excluded from the representation that respondents form of the target category. In the present case, these operations were a function of the specific content of the knowledge questions asked.

However, context effects can only be detected if the majority of the sample shares the same evaluation of the information that comes to mind. Suppose, for example, that only half the respondents thought highly of Richard von Weizsäcker. In that case, his inclusion in the representation of the CDU would have resulted in more favorable judgments for some respondents but less favorable judgments for others. Whereas each of these effects would reflect an assimilation of the general judgment to the evaluation of Richard von Weizsäcker at the theoretical level, these effects could have canceled one another, resulting in the apparent absence of context effects in the sample as a whole. Schwarz, Strack, and Mai (1991) observed such a finding in another study. In this study, thinking about one's marriage increased life satisfaction for happily married respondents but decreased life satisfaction for unhappily married ones, resulting in the absence of a context effect in the sample as a whole. It is therefore important to keep in mind that context effects are conditional (see Smith, 1992). For each respondent, the impact of the same general cognitive process depends on the implications of the specific information that is brought to the particular respondent's mind. Unless the conditional character of context effects is acknowledged in analyses, we may erroneously conclude that none were observed.

The implications of the conditionality of context effects for data analysis are often overlooked. Most important, the conditionality implies that any analysis of the impact of Question 1 on the answers given to Question 2 has to take respondents' substantive answers to Question 1 into account. Simply assessing the answers to Question 2 as a function of whether Question 1 preceded it or not is insufficient. Rather, we have to analyze the answers to Question 2 as a function of the placement of Question 1 and the answers given to it. This may be done by appropriate cross-tabulations or by a comparison of the correlation between both questions under both order conditions. In many cases, such a conditionality analysis reveals context effects where none are observed if only the overall means are compared.

The Number of Preceding Questions

According to the inclusion/exclusion model, the impact of a given piece of information depends on the amount and extremity of competing information used in constructing a representation. Accordingly, adding a piece of information to the representation of the target results in a larger assimilation effect when the representation contains little rather than a lot of other information.

Consistent with this assumption, Schwarz, Strack, and Mai (1991) observed that answering a marital satisfaction question before answering a question about one's general life satisfaction significantly increased the correlation of both measures from $r = .32$ (in the general-specific order) to $r = .67$ (in the specific-general order) when the marital satisfaction question was the only domain satisfaction question asked. However, the increase in correlation was less pronounced—and not significant—when three specific life domains (work, leisure, and marriage) were addressed prior to the general life satisfaction question, resulting in a correlation of $r = .42$ for marital and general satisfaction. Thus, the impact of information bearing on respondents' marriages was less pronounced as other information relevant to their lives became more accessible.

These effects, however, can be quite sensitive to what appear to be small changes in the number or content of previous questions. For example, Tourangeau, Rasinski, and Bradburn (1992) found what appeared on the surface to be a nonreplication of the results of Schwarz, Strack, and Mai (1991). Upon closer examination of the experimental conditions in the two studies, however, it was found that Tourangeau, Rasinski, and Bradburn (1992) asked about marital status before asking either the general or marital happiness question whereas Schwarz, Strack, and Mai asked about marriage and dating relationships without assessing respondents' marital status first. The question on marital status appears to have temporarily heightened the accessibility of the respondents' marriage in the Tourangeau, Rasinski, and Bradburn (1992) study and, as a result, explicitly asking the marital satisfaction question first did not further increase the impact of marriage-related information on general life satisfaction. In both studies, however, when the items were explicitly linked by a joint introduction, making it clear that they were related to one another, the results were similar: there was a lower correlation when the marital happiness question came before the general happiness question. This is a classic contrast effect, which we will discuss in more detail below.

The Generality of the Target Question

One of the most important determinants of assimilation versus contrast effects in survey practice is probably the generality of the target question. A question about the trustworthiness of politicians, for example, could refer to all politicians in the United States, to politicians of one party, or to a specific politician. In psychological terms, these questions would address target categories of different widths. The first question pertains to a wide category whereas the second question pertains to a narrower one. The last question, pertaining to one politician alone, is the narrowest and would not allow the inclusion of any other politician; the given person would make up a category by himself or herself. How would this differential category width affect the emergence of context effects?

Let us suppose, for example, that a preceding question asks respondents to recall some politicians who were involved in a scandal, rendering these politicians highly accessible. According to the model, the politicians involved in the scandal are members of the general category "politicians" and are therefore likely to be included in the temporary representation that respondents form of that category. If so, their evaluation of the trustworthiness of politicians in general should decrease, reflecting an assimilation effect. But the politicians who were involved in the scandal could not be included in the representation formed of the narrow category of one politician. Hence, the scandal-ridden politicians might now serve as a standard of comparison or scale anchor, resulting in a contrast effect.

A laboratory experiment with German respondents (Schwarz & Bless, 1992b) confirmed this prediction. (See Table 5.2.) Respondents who were asked to name two politicians involved in a well-known scandal subsequently reported less trust in German politicians in general. However, the same context question increased the reported trustworthiness of each of three specific politicians, although these exemplars were not perceived as particularly trustworthy to begin with. This pattern of findings suggests that the scandal-ridden politicians could be included in the representation formed of "German politicians" in general but not in the representation of any specific person. Since a question about "scandals" did tap the trustworthiness dimension to which the subsequent ratings

TABLE 5.2 Evaluation of trustworthiness of politicians.

	Scandal question	
	Not asked	*Asked*
Target		
Politicians in general	5.0	3.4
Specific exemplars	4.9	5.6

Source: Adapted from Schwarz and Bless, 1992b.
Note: $N = 8$ per condition; 1 = not at all trustworthy and 11 = very trustworthy.

pertained, the scandal-ridden politicians were used to construct the standard, creating a contrast effect.

In general, the inclusion/exclusion model predicts that assimilation effects are more likely to emerge when the target category is more inclusive. Accordingly, *general questions*, which assess respondents' opinions about a wide target category and allow for the inclusion of a variety of information, should be most likely to show assimilation effects. But *specific questions*, which assess respondents' opinions about a narrowly defined target, should be more likely to show contrast effects. The reason for this is that it is more likely that information that comes to mind can be included in a representation of a global target than of a specific target.

This assumption accounts for a variety of different observations. For example, Americans typically report low trust in congressmen in general but high trust in their own representative (see Erikson, Luttberg, & Tedin, 1988). Similarly, Americans are likely to endorse capital punishment in general but less likely to endorse it in any specific case (Ellsworth & Gross, 1994). Findings of this type suggest that extreme exemplars are likely to come to mind when respondents are asked a general question, presumably because extreme exemplars receive more attention in public discourse. These exemplars are included in the representation formed of a wide target category, resulting in assimilation effects on the general question. However, they cannot be included in the representation of the narrower target category, resulting in contrast effects on the specific question. As a result, the recalled examples of untrustworthy congressmen, for example, reduce trust in congressmen in general but increase trust in one's own Representative. Hence, apparent general-specific paradoxes of this type are actually a reflection of the impact of category width on the operation of inclusion/exclusion processes.

The Spacing of Questionnaire Items

The impact of item spacing in a questionnaire is likely to differ depending on the stage at which the respective context effect arises. At the question comprehension stage, respondents may deliberately turn to preceding items to make sense of a subsequent ambiguous question, as we have seen in Chapters Three and Four (for example, Strack, Schwarz, & Wänke, 1991). Adjacent items are more likely to be considered relevant than items placed at further distance from one another, reflecting a natural assumption that blocks of questions bear on related issues, much as they would during ordinary conversations. Moreover, respondents may lose track of more distant items, reducing the likelihood of their use. As a result, the impact of a context item at the question comprehension stage is likely to decrease as the number of intervening items increases.

At the judgment stage, however, the relationship between item spacing and context effects is more complicated. On the one hand, a large number of intervening items may interfere with recall of the primed information, again eliminating the emergence of a context effect (for example, Tourangeau, Rasinski, Bradburn, & D'Andrade, 1989a, 1989b). If the primed information still comes to mind, however, the spacing of the items may determine the use of the information, that is, its inclusion in or exclusion from the representation of the target. This is likely to be the case for two related reasons. First, psychological experiments have shown that respondents exclude information that comes to mind for what they assume to be the "wrong" reason. For example, Lombardi, Higgins, and Bargh (1987) observed that priming effects in a person perception task

were obtained only when respondents were not aware of the priming episode (see also Strack, Schwarz, Bless, Kübler, & Wänke, 1993). If respondents are aware that the information that comes to mind may do so only because it was triggered by a preceding question they may disregard (exclude) it for that reason (Martin, 1986). As a second possibility, conversational norms may induce respondents to ignore information they have already provided in response to a specific question when they are later asked a more general one because conversational norms request us to provide information that is "new" to the recipient rather than reiterate information already given, as we saw in Chapter Three. (See Schwarz, 1994; Strack & Schwarz, 1992; Tourangeau & Rasinski, 1988, for more detailed discussions.) Both of these processes may be influenced by the spacing of items and by introductions to blocks of related items (to be addressed in the following section).

A study by Ottati, Riggle, Wyer, Schwarz, and Kuklinski (1989) illustrates the impact of item spacing on inclusion/exclusion operations. They asked respondents to report their agreement with general and specific statements pertaining to civil liberties. For example, a general statement read, "Citizens should have the right to speak freely in public." In one condition, this general statement was preceded by a specific statement that pertained to a favorable or unfavorable group, that is, "The Parent-Teacher Association (or the Ku Klux Klan, respectively) should have the right to speak freely in public." As expected, respondents expressed a more favorable attitude toward the general statement if it was preceded by a specific one that pertained to a favorable group. However, this assimilation effect was only obtained when the items were separated by eight filler items. If the items were presented one after the other, a contrast effect emerged. The latter finding presumably reflects the exclusion of the primed information as a function of conversational norms or of awareness of the preceding item's possible influence.

Thus, assimilation effects are likely to emerge when questions are separated whereas contrast effects are likely to emerge when they are consecutive. However, as the number of filler items (often called "buffer items" by survey researchers) increases, they may interfere with the recall of previously activated information, eliminating the basis for either type of context effect (see Tourangeau, 1992; Tourangeau & Rasinski, 1988, for reviews). Hence, the number of intervening questions may determine the emergence of assimilation or of contrast effects as well as the absence of both.

Introductions to Item Blocks and Graphical Layout

As discussed in Chapter Three, in general conversation communicators are expected to avoid redundancy (Grice, 1975). In psycholinguistics this is known as the "given-new contract," which requires speakers to provide information that is "new" rather than to reiterate information already "given" (Haviland & Clark, 1974). Several studies indicate that this conversational norm of nonredundancy is evoked when related questions are perceived as belonging to the same conversational context (Schwarz, Strack, & Mai, 1991; Strack, Martin, & Schwarz, 1988; Strack, Schwarz, and Wänke, 1991). If the questions follow a part-whole format, using the information that has already been provided in response to a specific ("part") question in answer to a subsequent more general ("whole") question would violate this conversational norm. Variables that evoke the norm are introductions to a block of related items and the graphical presentation of self-administered questionnaires.

For example, in a study addressed earlier, Schwarz, Strack, and Mai (1991; see also Tourangeau, Rasinski, & Bradburn, 1992) asked respondents to report their marital satisfaction and their general life satisfaction. When the marital satisfaction question was asked as the last question on one page of the questionnaire and the general question as the first question on the next, happily married respondents reported higher general satisfaction and unhappily married respondents reported lower general satisfaction than when the general question came first, reflecting an assimilation effect. When both questions were introduced by a joint lead-in, thus assigning them to the same conversational context, respondents excluded the information that they reported in response to the specific question, resulting in contrast effects on the general one.

A related study (Schwarz & Hippler, unpublished data) pertaining to the graphical presentation of self-administered questionnaires provided a conceptual replication of this effect. Specifically, presenting marital satisfaction and life satisfaction as two separate questions, each framed by a black box, resulted in assimilation effects. Presenting the two questions in a single box—thus emphasizing their relatedness—elicited contrast effects.

These and similar findings (see Schwarz, 1994; Strack & Schwarz, 1992) demonstrate that conversational norms can trigger the exclusion of information that has already been provided from the cognitive representation formed for answering a subsequent question. Variables that can elicit the application of the conversational norm of nonredundancy include lead-ins to blocks of items as well as the graphical layout of self-administered questionnaires.

Implications for Attitude Theory

As the findings reviewed in this and the preceding chapter illustrate, attitude reports are subject to pronounced context effects. This fact is not surprising when we conceptualize attitude reports to reflect complex judgmental processes, as we do throughout this book. This judgmental perspective, however, is sometimes disputed.

The term *attitude* is usually used to refer to "a psychological tendency that is expressed by evaluating a particular entity with some degree of favor or disfavor" (Eagly & Chaiken, 1993, p. 1). Traditionally, attitudes were assumed to have a cognitive component, referring to beliefs about the attitude object; an affective component, referring to the evaluation of the object; and a conative component, referring to behavior toward the object (see Eagly & Chaiken, 1993, for a comprehensive review). This classic tripartite definition has been abandoned by most researchers primarily because the three components have been found to be only weakly related, rendering it difficult to think of them as part and parcel of the same thing. In fact, what is usually assessed in surveys is only the affective component of attitudes, as reflected in evaluative judgments. For this reason, we prefer the term *evaluative judgments* when we refer to attitude measures in the present book.

When we consider attitude measures to reflect evaluative judgments, their context dependency is not surprising: after all, human judgment is always context-dependent, no matter if it pertains to psychophysical stimuli or to complex social issues. This position, however, is not shared by all researchers. Rather, some have argued that people have well-formed attitudes with regard to some objects and that such well-formed attitudes are unlikely to be affected by the immediate context in which they are assessed. From this perspective, the emergence of context effects suggests that we are assessing "nonattitudes" (Converse, 1964), that is, responses to objects for which people do not have well-formed or "crystallized" attitudes.

In cognitive social psychology, these perspectives are often referred to as the "file-drawer" versus the "construal" model of attitudes (see Wilson & Hodges, 1992, for a recent discussion). According to the file-drawer model, well-formed attitudes pertaining to some objects are stored in memory and can be "looked up" when a person is asked to answer an attitude question. Only when this search is unsuccessful do we need to form a judgment on the spot. But according to the construal model, our mental file-drawer does not contain "attitudes" at all and we always need to form a judgment based on information that is chronically or temporarily accessible at the time. As an intermediate position, one may assume that the accessible information sometimes includes a previously formed judgment, which may enter the computation of a new judgment as another piece of information that simplifies the judgmental process.

In another take on the issue, Tourangeau and colleagues (see Tourangeau, 1992, for a comprehensive summary) suggested that we think of attitudes as memory structures that include valenced pieces of information. According to this theory, attitude reports usually require the computation of a judgment but the attitude concept is "saved" by redefining attitudes as memory structures. Thus, the information stored in memory constitutes a respondent's attitude but reports of this attitude may vary as a function of context, reflecting that the context influences which pieces of the memory structure are brought to mind at any given point. Hence, individuals "have" attitudes but their reports vary as a function of context. This theory essentially coincides with the construal model except that the term *attitude* is maintained by redefining the concept as pertaining to chronically accessible information.

Of course, these distinctions are only useful to the extent that they generate different predictions for observable phenomena. In this last section, we argue that deriving such crucial predictions is far more difficult than one might expect. To make this point, we adopt a strong version of a construal model, assuming for the sake of the argument that respondents always need to compute a judgment from scratch. Starting from this assumption, we apply the principles of the inclusion/exclusion model to findings that are usually supposed to support the assumption that individuals have attitudes on some topics, but perhaps not on others. Throughout, we will show that these findings are as compatible with a construal model as they are with the classic attitude model, rendering the debate futile for the purpose of attitude measurement in surveys.

The Stability of Attitude Reports over Time

An issue that we have already addressed is the stability of attitude reports over time. Presumably, obtaining similar reports at different points in time suggests that respondents have an attitude toward an object that they can report with some accuracy. As we discussed in considerable detail previously, however, construal models predict similar responses at different points in time under conditions that are likely to elicit only small context effects. Accordingly, construal models are compatible with the observation of stability in attitude reports and specify the conditions under which such stability should be observed, rendering the approach clearly testable. In contrast, the conclusion that individuals must have a well-formed attitude because their reports are stable over time is utterly circular in the absence of other evidence. One such set of evidence pertains to measures of attitude strength.

Attitude Strength

Several researchers have suggested that attitudes vary in their degree of "strength," "centrality," or "crystallization" (see Krosnick & Abelson, 1992, for a review). These concepts have been found to be difficult to operationalize (see Scott, 1966) and researchers have used a variety of indicators to assess attitude strength, including the intensity of respondents' feelings about the object, the certainty with which they report holding the attitude, and the importance they ascribe to it. Unfortunately, the various measures of attitude strength are only weakly related to one another. Moreover, the widely shared hypothesis that context effects in survey measurement "are greater in the case of weaker attitudes has clearly been disconfirmed" (Krosnick & Abelson, 1992, p. 193). In the most comprehensive test of this hypothesis, based on more than a dozen experiments and different measures of attitude strength, Krosnick and Schuman (1988) found no support for it except for the not-surprising finding that respondents with a weak attitude are more likely to choose a middle alternative. But given the high plausibility of the hypothesis and its long tradition in survey theorizing (dating back at least to Blankenship, 1943; Cantril, 1944; Katz, 1960), it is likely to remain popular.

However, attitude strength has proved to be important in other domains of research. Most important, strongly held attitudes have been found to be more stable over time and less likely to change in response to persuasive messages. Moreover, they are better predictors of behavior than weak attitudes (see Krosnick & Abelson, 1992, for a review). Once again, however, a construal model allows the same predictions. To the extent that we are likely to think more, and more often, about topics that are important to us, a larger

amount of information would be chronically accessible. Increased chronic accessibility, in turn, would result in higher stability over time and would decrease the likelihood of arriving at a different judgment when a few new pieces of information are added to the representation of the target. Moreover, an individual would be likely to draw on a similar set of information when asked to report a judgment and when faced with a behavioral decision, resulting in a higher consistency between the judgment and the behavior. Accordingly, a construal model would arrive at the same predictions if we assume that people think more about issues that are important to them, thus rendering more information chronically accessible, a plausible assumption. As a result, the findings of the attitude strength literature do not necessarily reflect that the processes underlying reports of "strong" attitudes differ from the processes underlying reports of "weak" attitudes.

Attitude Accessibility

Another body of research (for reviews see Bassili, 1995; Dovidio & Fazio, 1992; Fazio, 1990) suggests that some attitudes are more accessible than others, as reflected in respondents' reaction times. Presumably, a quick response to an attitude question reflects that a previously formed evaluation was accessible in memory whereas a slow response reflects computation of an evaluation. Several studies have found that highly accessible attitudes, as inferred from quick answers, are more stable over time and are better predictors of behavior (see Fazio, 1990).

Unfortunately, reaction time measures do not tell us which stage of the judgment process produces a fast or a slow response. A fast reaction may reflect the retrieval of a highly accessible previous judgment or the high speed of a current computation. Let us suppose, for example, that all information that comes to mind is evaluatively consistent. It would be easier to arrive at an overall judgment with this information than with inconsistent information. In the latter case, we would need to determine which weight we want to give to positive or negative features to arrive at an overall judgment. This would presumably take time, resulting in a slower response. But differences in reaction time might reflect differences in the evaluative consistency of the accessible information rather than the accessibility of a previously formed judgment.

Making this assumption, which is amenable to empirical testing, construal models would again arrive at the same predictions. To the extent that different pieces of chronically accessible information have similar implications, retrieving different pieces at different times would not result in different judgments. Moreover, retrieving different pieces when one makes a judgment and when one makes a behavioral decision would still result in a high consistency between the judgment and the behavior. It is only when different pieces of information have opposite implications that we should see low stability over time and low attitude-behavior consistency. Integrating the implications of these different pieces of information, however, would take time, resulting in the observed relationship between response time and stability or attitude-behavior consistency. Hence, the observed relationship does not necessarily reflect differences in the accessibility of existing attitudes, but may reflect differences in a mental construal process.

Concluding Thoughts

Our conclusion from these conjectures is not that the literature on attitude strength or attitude accessibility is necessarily mistaken. At present, we are not aware of data that bear on our conjectures. We simply note that the available findings are potentially compatible with construal models if one makes some rather plausible additional assumptions. In our reading, this suggests that asking whether people "have" attitudes is futile as far as survey measurement is concerned. We do not necessarily need different models for the measurement of attitudes and nonattitudes to account for the available findings. Rather, we can use a construal model of the type described (Schwarz & Bless, 1992a) to conceptualize the processes underlying the measurement of evaluative judgments, no matter whether they pertain to "attitudes" or "nonattitudes." Such a model has several advantages: it can account for stability as well as change; it predicts the conditions under which context effects are or are not likely to emerge; it predicts their direction, size, and generalization across targets; and it allows for the conceptualization of differences between respondents (such as expertise, attitude strength, and so on) as well as of questionnaire variables within a single conceptual framework. For the time being, we consider such an approach to be most likely to advance our understanding of survey measurement, although future research may teach us otherwise.

Summary

The inclusion/exclusion model provides a conceptual framework that allows predictions regarding the emergence, direction, size, and generalization of context effects in attitude measurement. Most important, the model holds that any variable that influences the categorization of information that comes to mind is likely to moderate the emergence of assimilation or contrast effects. Schwarz and Bless (1992a) provide a comprehensive review of a host of different variables that have been studied in psychological research. Most relevant to questionnaire construction are the content and number of preceding questions, the width of the target category, the spacing of items in a questionnaire, the lead-in to blocks of related questions, and the graphical presentation of self-administered questionnaires. Moreover, the model allows for the conceptualization of respondent variables such as expertise, motivation, and cognitive ability within the same conceptual framework and offers a number of straightforward predictions about the impact of the data-collection mode. Although some of the key predictions of the model have received considerable experimental support, the evidence bearing on others is rather limited. In addition, future research is likely to uncover other variables that may elicit inclusion/exclusion operations. We hope, however, that the present conceptualization provides a fruitful heuristic framework for future work in this area.